

SOHAM CHINCHALKAR

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EDUCATION

Master of Science in Information Systems

Aug 2024 - May 2026

Arizona State University

GPA: 4.0 / 4.0

- Coursework: Advanced Data Analytics, Deep Learning, Statistics, Adv. DBMS, Data in the Cloud, Computer vision, NLP

Bachelor of Engineering in Computer Engineering

Aug 2020 - Jun 2024

Savitribai Phule Pune University

GPA: 3.9 / 4.0

- Coursework: Data Structures and Algorithms, OOP, Networks, AI, Machine Learning, Data Science

SKILLS

- Languages & Frameworks: Python, Flask, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Java, SQL, SAS
- APIs & Services: REST APIs, Microservices, Data Modeling, Distributed Systems, ETL Pipelines
- Databases: MySQL, PostgreSQL, MongoDB, Snowflake, Neo4j, NoSQL, Relational DBMS concepts
- Cloud & DevOps: AWS (S3, Lambda), Azure, GCP, Docker, Kubernetes, Git, Azure DevOps, CI/CD
- Methodologies: Agile, OOP, Design Patterns, Documentation, Problem Solving

PROFESSIONAL EXPERIENCE

Data Scientist, Rose AI, New York, US

May 2025 – Aug 2025

- Designed and deployed Python-based services and APIs to process terabyte-scale financial datasets, ensuring data integrity and high performance.
- Built and maintained Flask microservices supporting internal data access layers, improving usability across teams.
- Architected a multi-tier data mart with distributed caching for low-latency analytics, aligning with enterprise compliance and scalability standards.
- Automated validation and monitoring pipelines using Pandas, reducing turnaround time on data quality checks by 28%.
- Collaborated in an Agile squad with PMs and engineers, expanding product offerings based on client and business requirements.
- Supported production systems by troubleshooting ETL and API failures, ensuring continuity of mission-critical analytics pipelines.

Data Engineer, Virtual Galaxy Infotech, Nagpur, India

Feb 2023 – May 2023

- Developed ETL workflows and APIs in Python and optimized SQL queries for 5 major banks, reducing reporting time by 30%.
- Resolved 50+ production database issues, improving reliability and reducing downtime by 15%.
- Partnered with cross-functional teams in Agile sprints (JIRA) to enhance functionality and improve performance.

Graduate Teaching Assistant, W. P. Carey School of Business, ASU

Aug 2024 – May 2025

- Delivered sessions on Python, SQL, and data modeling for 200+ students and professionals.
- Built interactive dashboards with live SQL + APIs, improving reporting and student engagement by 25%.
- Authored technical documentation and tutorials, enhancing operational continuity and user experience.

RESEARCH PUBLICATIONS AND PROJECTS

Financial Fraud Detection System Using Anti-Benford Subgraphs and ML Algorithms

Oct 2023 – Jan 2024

IEEE AIKIE 2023 | Scopus-Indexed Conference | DOI: [10.1109/AIKIE60097.2023.10390325](#)

- Developed an unsupervised fraud detection system using Anti-Benford graph mining and machine learning, achieving 94.83% anomaly detection accuracy on financial transaction networks.
- Developed an ML-powered fraud detection system with Pandas, NumPy, Scikit-learn, achieving 94.8% anomaly detection accuracy.
- Patent Published** – Anti-Benford & Louvain-Based Financial Fraud Detection, App. No. 202521025821, India, Apr 2025

Keylogger Detection System Using ML Algorithms and Dendritic Cell Algorithm

Jun 2023 – Feb 2024

First Author | Published in Scopus-Indexed Journal, Revue d'Intelligence Artificielle (IETA) | DOI: [10.18280/ria.380128](#)

- Built and productionized Python ML pipelines with 99.8% detection accuracy; implemented systematic testing and documentation.
- Developed end-to-end ML pipelines in Python using NumPy, Pandas, and Scikit-learn, improving detection precision through custom feature engineering and model tuning.
- Filed a patent for the designed system (App. No. 202421040728, May 2024).**

An LLM Augmented Knowledge Graph System for E-commerce Insights [View Project](#)

Feb 2025 – Apr 2025

- Built a RAG-based system combining LLMs and Neo4j knowledge graphs to extract entities and relationships from structured (100K+ Amazon records) and unstructured (50K+ chat logs) data, enabling semantic NL querying and dynamic graph construction.
- Achieved 98% accuracy in generating relevant, explainable outputs by translating NL queries to Cypher via LLMs and enriching the graph with inferred patterns for actionable insights like product trends and recommendations.

NYC Airbnb Trends & Host Analysis – Interactive Tableau Dashboard [View Dashboard](#)

Apr 2025 – May 2025

- Built an interactive Tableau dashboard analysing 48K+ listings with advanced features like geospatial mapping and drill-downs, highlighting top neighborhoods with 3,000+ listings and review scores above 4.8; applied data-ink ratio principles to reveal that 70%+ of listings are managed by multi-property hosts, driving strategic insights on pricing, guest satisfaction, and host behavior.

ACHIEVEMENTS

- Engineering Graduate Fellowship from Fulton Schools of Engineering.
- 2 Merit based scholarships from Arizona State University.